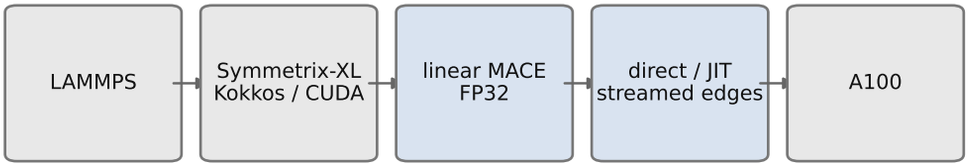
Symmetrix-Polar architecture evolution and measured speed (2026-10-22)

(a) Baseline: paper / upstream

linear MACE, FP32, no global SCF, no exact Ewald



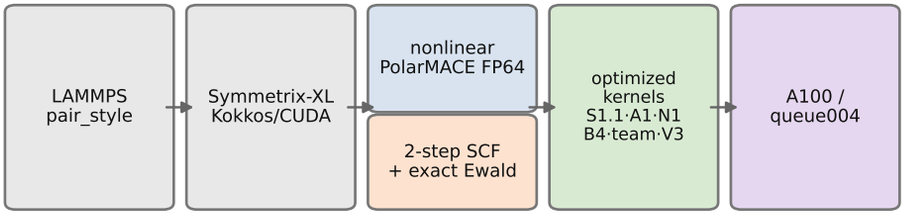
Measured on the paper's own workload:

4.50 µs/atom/step at 5,184 atoms

This is the matched streaming baseline; it does not run the polar model.

(b) Current: PolarMACE engine

same trained model; different execution path



Measured end-to-end:

145.75 µs/atom/step at 1,296 atoms

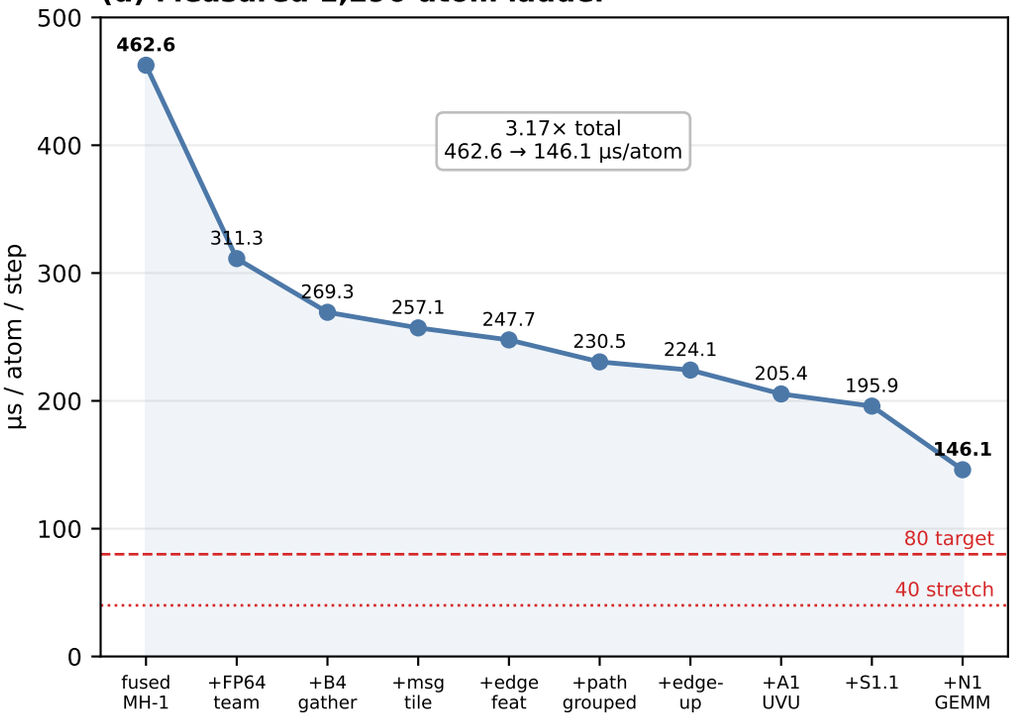
140.91 µs/atom at 5,184 atoms; parity with local A100

(c) Implementation changes (model unchanged)

Change	Measured effect
S1.1 node-contiguous streamed-edge tiles + fused reverse	**253.84 ms @1,296; 8.55 GiB**
A1 UVU 32×16 tile, chunked weight staging	**290.20 → 266.21 ms (−8.26%)**
N1 grouped FP64 path-GEMM (6 tables, 20 planes)	**255.35 → 189.30 ms (−25.87%)**
Team MH-1 + B4 + message/edge/path tiles	**462.6 → 224.1 µs/atom**
V3 prepared path-plan cache	**~24 D2H syncs/eval removed**
queue004 packaged offload (same binary/model)	**145.75 = local within 0.2%**

Weights, equations, FP64, exact Ewald, and the LAMMPS interface are unchanged.

(d) Measured 1,296-atom ladder



Sources: docs/results/OPTIMIZATION_LEDGER.md; docs/results/LATEST_SPEED_AND_PLAN_2026-10-22.md. The polar model's weights and equations are unchanged; all gains are execution changes.